

NEW CENTRAL q -BINOMIAL IDENTITIES

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ABSTRACT. We establish several new series evaluations involving the central q -binomial coefficients, with the inspiration coming from earlier work by Vignat and one of the authors on the limiting case at $q \rightarrow 1$.

1. INTRODUCTION

The *central binomial coefficients* $\binom{2k}{k}$ play an important role in the theory of infinite series. A classic example arises from the Maclaurin expansion of the inverse trigonometric function $\arcsin z$, asserting that [5, p. 121, eq. (4.24.1)]:

$$\arcsin z = \sum_{k \geq 0} \binom{2k}{k} \frac{z^{2k+1}}{4^k(2k+1)},$$

for $z \in \mathbb{C}$ with $|z| \leq 1$. This produces a family of series for π including

$$\frac{\pi}{2} = \sum_{k \geq 0} \binom{2k}{k} \frac{1}{4^k(2k+1)}.$$

Along this line, Vignat and one of the authors [2, Theorem 2.1] established the following generalization for $\alpha \in \mathbb{C} \setminus \{-1, -3, -5, \dots\}$,

$$\sum_{k \geq 0} \binom{2k}{k} \frac{1}{4^k(2k+1+\alpha)} = \frac{\Gamma(\frac{3}{2}) \Gamma(\frac{\alpha+1}{2})}{\Gamma(\frac{\alpha+2}{2})} = \frac{\sqrt{\pi}}{2} \frac{\Gamma(\frac{\alpha+1}{2})}{\Gamma(\frac{\alpha+2}{2})}, \quad (1.1)$$

where $\Gamma(z)$ is the *gamma function*.

As is often the case, it is of general interest to consider q -analogs of these infinite series evaluations. We recall that the binomial coefficient $\binom{n}{m}$ is the specialization at $q \rightarrow 1$ of the q -binomial coefficient

$$\begin{bmatrix} n \\ m \end{bmatrix}_q := \begin{cases} \frac{(q; q)_n}{(q; q)_m (q; q)_{n-m}}, & \text{if } 0 \leq m \leq n; \\ 0, & \text{otherwise,} \end{cases}$$

where the q -Pochhammer symbols are defined for $N \in \mathbb{N} \cup \{\infty\}$,

$$(a; q)_N := \prod_{j=0}^{N-1} (1 - aq^j),$$

$$(a_1, a_2, \dots, a_l; q)_N := (a_1; q)_N (a_2; q)_N \cdots (a_l; q)_N.$$

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The main objective of this paper is to investigate series evaluations involving the central q -binomial coefficients.

Although shown by an elegant technique of integration in [2], the series in (1.1) can be more universally understood as a ${}_2F_1$ hypergeometric series [1, p. 62, eq. (2.1.2)]. Namely,

$$\sum_{k \geq 0} \binom{2k}{k} \frac{1}{4^k(2k+1+\alpha)} = \frac{1}{\alpha+1} {}_2F_1 \left(\frac{1}{2}, \frac{\alpha+1}{2}; 1 \right).$$

This then can be evaluated to the gamma-quotient in (1.1) by the *Gauß summation formula* [1, p. 66, Theorem 2.2.2]. It is known that the ${}_2F_1$ series can be generalized to the *basic hypergeometric series*

$${}_r\phi_s \left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z \right) := \sum_{k \geq 0} \left((-1)^k q^{\binom{k}{2}} \right)^{1+s-r} \frac{(a_1, \dots, a_r; q)_k}{(b_1, \dots, b_s; q)_k} z^k.$$

In view of the q -*Gauß summation* [3, p. 354, eq. (II.8)]:

$${}_2\phi_1 \left(\begin{matrix} a, b \\ c \end{matrix}; q, \frac{c}{ab} \right) = \frac{(c/a, c/b; q)_\infty}{(c, c/(ab); q)_\infty}, \quad (1.2)$$

it is not surprising to see the following q -analog of (1.1):

$$\begin{aligned} \sum_{k \geq 0} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [2k+1+\alpha]_q} &= \frac{1}{[\alpha+1]_q} {}_2\phi_1 \left(\begin{matrix} q, q^{\alpha+1} \\ q^{\alpha+3} \end{matrix}; q^2, q \right) \\ &= \frac{\Gamma_{q^2}(\frac{3}{2}) \Gamma_{q^2}(\frac{\alpha+1}{2})}{\Gamma_{q^2}(\frac{\alpha+2}{2})}, \end{aligned} \quad (1.3)$$

where we have adopted the q -integers

$$[n]_q := 1 + q + \dots + q^{n-1} = \frac{1 - q^n}{1 - q},$$

and the q -gamma function

$$\Gamma_q(z) := \frac{(q; q)_\infty}{(q^z; q)_\infty} (1 - q)^{1-z}.$$

What makes our study more interesting is the singular case of the series in (1.1). Removing the singular term, the following series evaluation connecting the *alternating harmonic numbers*

$$\bar{H}_n := \sum_{k=1}^n \frac{(-1)^{k-1}}{k}$$

was shown in [2, Theorem 3.1].

Theorem 1.1 (Dilcher–Vignat). *For all nonnegative integers m ,*

$$\sum_{\substack{k \geq 0 \\ k \neq m}} \binom{2k}{k} \frac{1}{4^k(k-m)} = \binom{2m}{m} \frac{\log 2 - \bar{H}_{2m}}{2^{2m-1}}. \quad (1.4)$$

In this work, we focus on the natural q -analog of the series in (1.4):

$$\sum_{\substack{k \geq 0 \\ k \neq m}} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}}. \quad (1.5)$$

Our main result is as follows.

Theorem 1.2. *For all nonnegative integers m ,*

$$\sum_{\substack{k \geq 0 \\ k \neq m}} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} = \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1+q)q^m}{(-q; q)_{2m}} \sum_{k \geq 2m+1} \frac{(-1)^{k-1} q^k}{[k]_q}. \quad (1.6)$$

Remark 1.3. Letting $q \rightarrow 1$, we in particular have

$$\lim_{q \rightarrow 1} \sum_{k \geq 2m+1} \frac{(-1)^{k-1} q^k}{[k]_q} = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} - \sum_{k=1}^{2m} \frac{(-1)^{k-1}}{k} = \log 2 - \bar{H}_{2m},$$

thereby recovering (1.4).

Outline of the paper. First in Section 2, we give a direct proof of our main result in Theorem 1.2. Meanwhile, we note that the series (1.5) consists of two components, one *infinite* and the other *finite*, and each of them should have independent interest. Thus, in Sections 3 and 4, we work on the two summations separately — combining the relations in Theorems 3.1 and 4.1 produces an alternative proof of Theorem 1.2, although this makes a detour in comparison with the one in Section 2. Finally, the evaluation for the finite component suggests some variants of Theorem 4.1, and we will discuss them in Section 5.

2. A DIRECT PROOF OF THEOREM 1.2

In this section, we provide a direct proof of Theorem 1.2. In doing so, we need a widely known trick of introducing auxiliary variables for series transformations. First, we note that

$$\begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{1}{(-q; q)_{2k}} = \frac{(q; q^2)_k}{(q^2; q^2)_k}. \quad (2.1)$$

Therefore,

$$\sum_{\substack{k \geq 0 \\ k \neq m}} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} = (1-q^2) \sum_{\substack{k \geq 0 \\ k \neq m}} \frac{(q; q^2)_k q^k}{(q^2; q^2)_k (1-q^{2k-2m})}.$$

Now let us define

$$F(a, q) := \frac{1}{1-aq^{-2m}} \left({}_2\phi_1 \left(\begin{matrix} q, aq^{-2m} \\ aq^{-2m+2}; q^2, q \end{matrix} \right) - \frac{(q, aq^{-2m}; q^2)_m q^m}{(q^2, aq^{-2m+2}; q^2)_m} \right).$$

In other words,

$$F(a, q) = \sum_{\substack{k \geq 0 \\ k \neq m}} \frac{(q; q^2)_k q^k}{(q^2; q^2)_k (1-aq^{2k-2m})},$$

so that

$$\sum_{\substack{k \geq 0 \\ k \neq m}} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} = (1-q^2) \lim_{a \rightarrow 1} F(a, q).$$

For the ${}_2\phi_1$ series in $F(a, q)$, the q -Gauß summation (1.2) gives us

$${}_2\phi_1 \left(\begin{matrix} q, aq^{-2m} \\ aq^{-2m+2}; q^2, q \end{matrix} \right) = \frac{(q^2, aq^{-2m+1}; q^2)_\infty}{(q, aq^{-2m+2}; q^2)_\infty}.$$

Thus,

$$\begin{aligned} F(a, q) &= \frac{1}{1-aq^{-2m}} \left(\frac{(q^2, aq^{-2m+1}; q^2)_\infty}{(q, aq^{-2m+2}; q^2)_\infty} - \frac{(q, aq^{-2m}; q^2)_m q^m}{(q^2, aq^{-2m+2}; q^2)_m} \right) \\ &= \frac{1}{1-a} \left(\frac{(q^2, aq; q^2)_\infty (aq^{-2m+1}; q^2)_m}{(q, aq^2; q^2)_\infty (aq^{-2m}; q^2)_m} - \frac{(q; q^2)_m q^m}{(q^2; q^2)_m} \right). \end{aligned}$$

Since

$$\lim_{a \rightarrow 1} \frac{(q^2, aq; q^2)_\infty (aq^{-2m+1}; q^2)_m}{(q, aq^2; q^2)_\infty (aq^{-2m}; q^2)_m} = \frac{(q^{-2m+1}; q^2)_m}{(q^{-2m}; q^2)_m} = \frac{(q; q^2)_m q^m}{(q^2; q^2)_m},$$

we may apply the L'Hospital rule to get

$$\begin{aligned} \lim_{a \rightarrow 1} F(a, q) &= -\frac{\partial}{\partial a} \bigg|_{a=1} \left(\frac{(q^2, aq; q^2)_\infty (aq^{-2m+1}; q^2)_m}{(q, aq^2; q^2)_\infty (aq^{-2m}; q^2)_m} - \frac{(q; q^2)_m q^m}{(q^2; q^2)_m} \right) \\ &= -\frac{(q^2; q^2)_\infty}{(q; q^2)_\infty} \frac{\partial}{\partial a} \bigg|_{a=1} \frac{(aq; q^2)_\infty (aq^{-2m+1}; q^2)_m}{(aq^2; q^2)_\infty (aq^{-2m}; q^2)_m}. \end{aligned}$$

Recall that for a generic function $f(a)$,

$$\frac{d}{da} f(a) = f(a) \frac{d}{da} \log f(a).$$

We then compute that

$$\begin{aligned} \lim_{a \rightarrow 1} F(a, q) &= -\frac{(q^{-2m+1}; q^2)_m}{(q^{-2m}; q^2)_m} \frac{\partial}{\partial a} \bigg|_{a=1} \left(\sum_{k \geq 1} \log \frac{1-aq^{2k-1}}{1-aq^{2k}} + \sum_{k=1}^m \log \frac{1-aq^{-2k+1}}{1-aq^{-2k}} \right) \\ &= \frac{(q; q^2)_m q^m}{(q^2; q^2)_m} \left(\sum_{k \geq 1} \frac{(-1)^{k-1} q^k}{1-q^k} + \sum_{k=1}^{2m} \frac{(-1)^{k-1} q^{-k}}{1-q^{-k}} \right) \\ &= \frac{(q; q^2)_m q^m}{(q^2; q^2)_m} \left(\sum_{k \geq 2m+1} \frac{(-1)^{k-1} q^k}{1-q^k} + \sum_{k=1}^{2m} (-1)^k \right) \\ &= \frac{(q; q^2)_m q^m}{(q^2; q^2)_m} \sum_{k \geq 2m+1} \frac{(-1)^{k-1} q^k}{1-q^k}. \end{aligned}$$

Finally,

$$\begin{aligned} \sum_{\substack{k \geq 0 \\ k \neq m}} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} &= (1-q^2) \lim_{a \rightarrow 1} F(a, q) \\ &= \frac{(q; q^2)_m (1+q) q^m}{(q^2; q^2)_m} \sum_{k \geq 2m+1} \frac{(-1)^{k-1} q^k}{[k]_q}, \end{aligned}$$

which is exactly the right-hand side of (1.6) after rewriting the prefactor in terms of the central q -binomial coefficient according to (2.1).

3. THE INFINITE COMPONENT

Now we move on to the infinite component in (1.5), which also exhibits a neat evaluation.

Theorem 3.1. *For all nonnegative integers m ,*

$$\begin{aligned} \sum_{k \geq m+1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} \\ = \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1+q) q^m}{(-q; q)_{2m}} \left(\sum_{k \geq 1} \frac{q^{2k-1}}{[2k-1]_q} - \sum_{k \geq 1} \frac{q^{2k+2m}}{[2k+2m]_q} \right). \end{aligned} \quad (3.1)$$

Proof. We have, by again applying (2.1) and then shifting the index $k \mapsto k+m$,

$$\begin{aligned} \sum_{k \geq m+1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} &= (1-q^2) \sum_{k \geq 1} \frac{(q; q^2)_{k+m} q^{k+m}}{(q^2; q^2)_{k+m} (1-q^{2k})} \\ &= \frac{(1-q^2) q^m (q; q^2)_m}{(q^2; q^2)_m} \sum_{k \geq 1} \frac{(q^{2m+1}; q^2)_k q^k}{(q^{2m+2}; q^2)_k (1-q^{2k})} \\ &= \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1-q^2) q^m}{(-q; q)_{2m}} \sum_{k \geq 1} \frac{(q^{2m+1}; q^2)_k q^k}{(q^{2m+2}; q^2)_k (1-q^{2k})}. \end{aligned}$$

For the moment, we introduce the auxiliary series

$$U(a, q) := \frac{1}{1-a} \left({}_2\phi_1 \left(\begin{matrix} a, q^{2m+1} \\ q^{2m+2} \end{matrix}; q^2, a^{-1}q \right) - 1 \right).$$

It is clear that

$$\begin{aligned} \lim_{a \rightarrow 1} U(a, q) &= \lim_{a \rightarrow 1} \frac{1}{1-a} \sum_{k \geq 1} \frac{(a, q^{2m+1}; q^2)_k (a^{-1}q)^k}{(q^2, q^{2m+2}; q^2)_k} \\ &= \left[\sum_{k \geq 1} \frac{(aq^2; q^2)_{k-1} (q^{2m+1}; q^2)_k (a^{-1}q)^k}{(q^2, q^{2m+2}; q^2)_k} \right]_{a=1} \\ &= \sum_{k \geq 1} \frac{(q^{2m+1}; q^2)_k q^k}{(q^{2m+2}; q^2)_k (1-q^{2k})}. \end{aligned}$$

Now applying the q -Gauß summation (1.2) to the ${}_2\phi_1$ series in $U(a, q)$, we obtain

$$U(a, q) = \frac{1}{1-a} \left(\frac{(q, a^{-1}q^{2m+2}; q^2)_\infty}{(a^{-1}q, q^{2m+2}; q^2)_\infty} - 1 \right).$$

By L'Hospital,

$$\begin{aligned} \lim_{a \rightarrow 1} U(a, q) &= - \frac{\partial}{\partial a} \bigg|_{a=1} \frac{(q, a^{-1}q^{2m+2}; q^2)_\infty}{(a^{-1}q, q^{2m+2}; q^2)_\infty} \\ &= - \left[\frac{(q, a^{-1}q^{2m+2}; q^2)_\infty}{(a^{-1}q, q^{2m+2}; q^2)_\infty} \sum_{k \geq 1} \frac{\partial}{\partial a} \log \frac{1 - a^{-1}q^{2k+2m}}{1 - a^{-1}q^{2k-1}} \right]_{a=1} \\ &= \sum_{k \geq 1} \frac{q^{2k-1}}{1 - q^{2k-1}} - \sum_{k \geq 1} \frac{q^{2k+2m}}{1 - q^{2k+2m}}. \end{aligned}$$

The desired relation then follows. $\square$

4. THE FINITE COMPONENT

Although the finite component in (1.5) can be computed by subtracting (3.1) from (1.6), we offer an independent proof for its own interest.

Theorem 4.1. *For all nonnegative integers m ,*

$$\sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} = - \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1+q)q^m}{(-q; q)_{2m}} \sum_{k=1}^m \frac{q^{2k-1}}{[2k-1]_q}. \quad (4.1)$$

Proof. Note that by (2.1),

$$\sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} = (1 - q^2) \sum_{k=0}^{m-1} \frac{(q; q^2)_k q^k}{(q^2; q^2)_k (1 - q^{2k-2m})}.$$

Let us define the auxiliary series

$$V(a, q) := \frac{1}{1 - q^{-2m}} \left({}_2\phi_1 \left(\begin{matrix} q, q^{-2m} \\ a^{-1}q^{-2m+2} \end{matrix}; q^2, a^{-1}q \right) - \frac{(q, q^{-2m}; q^2)_m (a^{-1}q)^m}{(q^2, a^{-1}q^{-2m+2}; q^2)_m} \right).$$

In particular,

$$V(a, q) = \sum_{k=0}^{m-1} \frac{(q, q^{-2m}; q^2)_k (a^{-1}q)^k}{(q^2, a^{-1}q^{-2m+2}; q^2)_k},$$

so that

$$\lim_{a \rightarrow 1} V(a, q) = \sum_{k=0}^{m-1} \frac{(q; q^2)_k q^k}{(q^2; q^2)_k (1 - q^{2k-2m})}.$$

Now for the terminating ${}_2\phi_1$ series in $V(a, q)$, we apply the *first q -Chu-Vandermonde summation* [3, p. 354, eq. (II.7)]:

$${}_2\phi_1 \left(\begin{matrix} a, q^{-N} \\ c \end{matrix}; q, \frac{cq^N}{a} \right) = \frac{(c/a; q)_N}{(c; q)_N}.$$

Thus,

$$\begin{aligned} V(a, q) &= \frac{1}{1 - q^{-2m}} \left(\frac{(a^{-1}q^{-2m+1}; q^2)_m}{(a^{-1}q^{-2m+2}; q^2)_m} - \frac{(q, q^{-2m}; q^2)_m (a^{-1}q)^m}{(q^2, a^{-1}q^{-2m+2}; q^2)_m} \right) \\ &= \frac{1}{1 - q^{-2m}} \left(\frac{(aq; q^2)_m q^{-m}}{(a; q^2)_m} - \frac{(q; q^2)_m q^{-m}}{(a; q^2)_m} \right) \\ &= -\frac{q^m}{1 - q^{2m}} \frac{(aq; q^2)_m - (q; q^2)_m}{(1 - a)(aq^2; q^2)_{m-1}}. \end{aligned}$$

It follows that

$$\begin{aligned} \lim_{a \rightarrow 1} V(a, q) &= -\frac{q^m}{1 - q^{2m}} \left(\lim_{a \rightarrow 1} \frac{1}{(aq^2; q^2)_{m-1}} \right) \left(\lim_{a \rightarrow 1} \frac{(aq; q^2)_m - (q; q^2)_m}{1 - a} \right) \\ &= -\frac{q^m}{(q^2; q^2)_m} \lim_{a \rightarrow 1} \frac{(aq; q^2)_m - (q; q^2)_m}{1 - a}. \end{aligned}$$

By L'Hospital,

$$\begin{aligned} \lim_{a \rightarrow 1} V(a, q) &= \frac{q^m}{(q^2; q^2)_m} \frac{\partial}{\partial a} \bigg|_{a=1} (aq; q^2)_m \\ &= \frac{q^m}{(q^2; q^2)_m} \left[(aq; q^2)_m \sum_{k=1}^m \frac{\partial}{\partial a} \log(1 - aq^{2k-1}) \right]_{a=1} \\ &= -\frac{(q; q^2)_m q^m}{(q^2; q^2)_m} \sum_{k=1}^m \frac{q^{2k-1}}{1 - q^{2k-1}}. \end{aligned}$$

Therefore,

$$\begin{aligned} \sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k - m]_{q^2}} &= (1 - q^2) \lim_{a \rightarrow 1} V(a, q) \\ &= -\begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1 - q^2) q^m}{(-q; q)_{2m}} \sum_{k=1}^m \frac{q^{2k-1}}{1 - q^{2k-1}}, \end{aligned}$$

as claimed. $\square$

5. FINITE VARIANTS

In the proof of Theorem 4.1, we have applied the first q -Chu–Vandermonde summation. Notably, this formula has a natural companion, which suggests the following variant of (4.1).

Theorem 5.1. *For all nonnegative integers m ,*

$$\sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^{2k}}{(-q; q)_{2k} [k - m]_{q^2}} = -\begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1 + q) q^{2m}}{(-q; q)_{2m}} \sum_{k=1}^m \frac{1}{[2k - 1]_q}. \quad (5.1)$$

Proof. In view of (2.1), this time we have

$$\sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^{2k}}{(-q; q)_{2k} [k - m]_{q^2}} = (1 - q^2) \sum_{k=0}^{m-1} \frac{(q; q^2)_k q^{2k}}{(q^2; q^2)_k (1 - q^{2k-2m})}.$$

Define the auxiliary series

$$\widehat{V}(a, q) := \frac{1}{1 - q^{-2m}} \left({}_2\phi_1 \left(\begin{matrix} q, q^{-2m} \\ a^{-1}q^{-2m+2}; q^2, q^2 \end{matrix} \right) - \frac{(q, q^{-2m}; q^2)_m q^{2m}}{(q^2, a^{-1}q^{-2m+2}; q^2)_m} \right),$$

which satisfies

$$\lim_{a \rightarrow 1} \widehat{V}(a, q) = \sum_{k=0}^{m-1} \frac{(q; q^2)_k q^{2k}}{(q^2; q^2)_k (1 - q^{2k-2m})}.$$

For the terminating ${}_2\phi_1$ series in $\widehat{V}(a, q)$, we need the *second q -Chu–Vandermonde summation* [3, p. 354, eq. (II.6)]:

$${}_2\phi_1 \left(\begin{matrix} a, q^{-N} \\ c \end{matrix}; q, q \right) = \frac{(c/a; q)_N a^N}{(c; q)_N}.$$

Thus,

$$\begin{aligned} \widehat{V}(a, q) &= \frac{1}{1 - q^{-2m}} \left(\frac{(a^{-1}q^{-2m+1}; q^2)_m q^m}{(a^{-1}q^{-2m+2}; q^2)_m} - \frac{(q, q^{-2m}; q^2)_m q^{2m}}{(q^2, a^{-1}q^{-2m+2}; q^2)_m} \right) \\ &= -\frac{q^{2m}}{1 - q^{2m}} \frac{(aq; q^2)_m - (q; q^2)_m a^m}{(1 - a)(aq^2; q^2)_{m-1}}, \end{aligned}$$

which further gives

$$\begin{aligned} \lim_{a \rightarrow 1} \widehat{V}(a, q) &= -\frac{q^{2m}}{(q^2; q^2)_m} \lim_{a \rightarrow 1} \frac{(aq; q^2)_m - (q; q^2)_m a^m}{1 - a} \\ &= \frac{(q; q^2)_m q^{2m}}{(q^2; q^2)_m} \left(-m + \sum_{k=1}^m \frac{\partial}{\partial a} \bigg|_{a=1} \log(1 - aq^{2k-1}) \right) \\ &= -\frac{(q; q^2)_m q^{2m}}{(q^2; q^2)_m} \sum_{k=1}^m \frac{1}{1 - q^{2k-1}}, \end{aligned}$$

where we have again applied the L'Hospital rule. The required identity then follows by substituting this limit into the first equation in the proof. $\square$

In light of Theorems 4.1 and 5.1, we derive an intriguing identity.

Corollary 5.2. *For all nonnegative integers m ,*

$$\sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} (1 + q^{k-m})} = \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{mq^{2m}}{(-q; q)_{2m}}. \quad (5.2)$$

Proof. The claimed result follows by dividing by q^m on both sides of (5.1) and then taking its difference with (4.1). $\square$

Remark 5.3. The limiting case of (5.2) at $q \rightarrow 1$ is

$$\sum_{k=0}^{m-1} \binom{2k}{k} \frac{1}{2^{2k+1}} = \binom{2m}{m} \frac{m}{4^m}. \quad (5.3)$$

This identity can be found in [4, p. 14, eq. (1.109)].

Besides (5.1), the q -Pfaff–Saalschütz summation produces a second variant of (4.1) in a *squared* manner.

Theorem 5.4. *For all nonnegative integers m ,*

$$\sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^{2k}}{(-q; q)_{2k}^2 [k-m]_{q^2}} = - \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1+q)q^{2m}}{(-q; q)_{2m}^2} \sum_{k=1}^m \frac{[4k-2]_q}{[2k-1]_q^2}. \quad (5.4)$$

Proof. Note that by (2.1),

$$\sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^{2k}}{(-q; q)_{2k}^2 [k-m]_{q^2}} = (1-q^2) \sum_{k=0}^{m-1} \frac{(q, q; q^2)_k q^{2k}}{(q^2, q^2; q^2)_k (1-q^{2k-2m})}.$$

Now we introduce the auxiliary series

$$W(a, q) := \frac{1}{1-q^{-2m}} \left({}_3\phi_2 \left(\begin{matrix} q, a^{-1}q, q^{-2m} \\ q^2, a^{-1}q^{-2m+2} \end{matrix}; q^2, q^2 \right) - \frac{(q, a^{-1}q, q^{-2m}; q^2)_m q^{2m}}{(q^2, q^2, a^{-1}q^{-2m+2}; q^2)_m} \right),$$

so that

$$\lim_{a \rightarrow 1} W(a, q) = \sum_{k=0}^{m-1} \frac{(q, q; q^2)_k q^{2k}}{(q^2, q^2; q^2)_k (1-q^{2k-2m})}.$$

For the terminating ${}_3\phi_2$ series in $W(a, q)$, we require the q -Pfaff–Saalschütz summation [3, p. 355, eq. (II.12)]:

$${}_3\phi_2 \left(\begin{matrix} a, b, q^{-N} \\ c, abc^{-1}q^{1-N} \end{matrix}; q, q \right) = \frac{(c/a, c/b; q)_N}{(c, c/(ab); q)_N}.$$

Then we arrive at

$$\begin{aligned} W(a, q) &= \frac{1}{1-q^{-2m}} \left(\frac{(q, aq; q^2)_m}{(q^2, a; q^2)_m} - \frac{(q, a^{-1}q, q^{-2m}; q^2)_m q^{2m}}{(q^2, q^2, a^{-1}q^{-2m+2}; q^2)_m} \right) \\ &= - \frac{q^{2m}}{1-q^{2m}} \frac{(q; q^2)_m}{(q^2; q^2)_m} \frac{(aq; q^2)_m - (a^{-1}q; q^2)_m a^m}{(1-a)(aq^2; q^2)_{m-1}}, \end{aligned}$$

thereby yielding

$$\begin{aligned} \lim_{a \rightarrow 1} W(a, q) &= - \frac{(q; q^2)_m q^{2m}}{(q^2; q^2)_m^2} \lim_{a \rightarrow 1} \frac{(aq; q^2)_m - (a^{-1}q; q^2)_m a^m}{1-a} \\ &= \frac{(q; q^2)_m^2 q^{2m}}{(q^2; q^2)_m^2} \sum_{k=1}^m \frac{\partial}{\partial a} \bigg|_{a=1} \log(1 - aq^{2k-1}) \\ &\quad - \frac{(q; q^2)_m^2 q^{2m}}{(q^2; q^2)_m^2} \left(m + \sum_{k=1}^m \frac{\partial}{\partial a} \bigg|_{a=1} \log(1 - a^{-1}q^{2k-1}) \right) \\ &= - \frac{(q; q^2)_m^2 q^{2m}}{(q^2; q^2)_m^2} \sum_{k=1}^m \frac{1+q^{2k-1}}{1-q^{2k-1}}. \end{aligned}$$

Finally, recalling that

$$\sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^{2k}}{(-q; q)_{2k}^2 [k-m]_{q^2}} = (1-q^2) \lim_{a \rightarrow 1} W(a, q),$$

the claimed result holds after a modest amount of elementary algebra. $\square$

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